Extra Practice 1: Squares, Square Roots and Triangle Basics

A. Skills

1. Find each square. Use your calculator for (d) to (f).

(a) 13^2 _____ (b) 7^2 _____ (c) 50^2 _____ (d) 0.4^2 _____ (e) 2.5^2 _____ (f) 4.2^2 _____

2. Find each square root. Do (a) to (d) without a calculator. For (e) to (h), use your calculator and round to the nearest tenth.

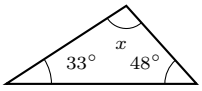
(a) $\sqrt{64}$ _____ (b) $\sqrt{9}$ _____ (c) $\sqrt{225}$ _____ (d) $\sqrt{900}$ _____

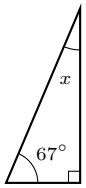
(e) $\sqrt{30}$ _____ (f) $\sqrt{7}$ _____ (g) $\sqrt{300}$ _____ (h) $\sqrt{0.5}$ _____

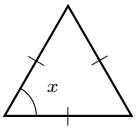
3. Each square root is between two whole numbers. Name them, and circle the one it is closer to.

(a) $\sqrt{14}$ is between ___ and ___ (b) $\sqrt{55}$ is between ___ and ___ (c) $\sqrt{99}$ is between ___ and ___

4. Find each missing angle.

(a) 

(b) 

(c) 

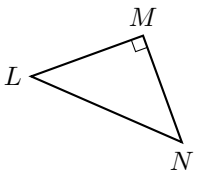
5. Name each triangle by its sides and by its angles.

(a) Its angles are 30°, 60° and 90°. _____

(b) Two sides are 10 cm long. Its angles are 80°, 50° and 50°. _____

(c) Its angles are 25°, 35° and 120°. _____

6. $\triangle LMN$ has its right angle at M.



(a) The hypotenuse is side _____, also called side _____.

(b) The legs are _____ and _____.

(c) The sides are 20 cm, 21 cm and 29 cm. Which length is LN?

_____

B. Applying

7. A square quilt block has an area of 324 cm^2.
(a) How long is each side of the block?

(b) A square quilt is made of 6 rows of 6 blocks. How long is each side of the quilt? What is its area?

8. Ravi floods a square backyard rink with an area of 160 m^2.
(a) How long is each side, to the nearest tenth of a metre?

(b) He puts boards all the way around the rink. How many metres of boards does he need, to the nearest tenth?

9. A ladder leans against the side of a shed. The ground is level and the wall is straight up. The ladder makes a 72° angle with the ground.
(a) Sketch the triangle. Mark the right angle and label the hypotenuse.

(b) What angle does the ladder make with the wall?

10. A roof truss is an isosceles triangle. Each of the two bottom angles is 22.5°. What is the angle at the peak? What kind of triangle is it by its angles?

11. Which number has a square root between 6 and 7?
A. 40 **B.** 30 **C.** 50 **D.** 6.5

12. Numerical Response A square tablecloth has an area of 2.89 m^2. Each side is _____ m long.

C. Stretch

13. Put these numbers in order from least to greatest without a calculator. Explain how you decided.

$\sqrt{45}$ 6.5 $\sqrt{40}$ 6.4

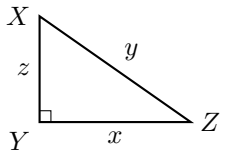
14. A right triangle has one acute angle that is twice as big as the other acute angle. Find all three angles. (Hint: guess and check, or think about what the two acute angles must add to.)

15. Find every whole number n so that $\sqrt{n}$ is between 8 and 9 *and* closer to 9 than to 8.

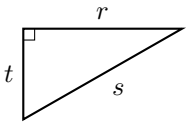
Extra Practice 2: The Pythagorean Theorem: Finding the Hypotenuse

A. Skills

1. For each right triangle, name the hypotenuse and write the Pythagorean theorem using its letters.



(a) Hypotenuse: _____



(b) Hypotenuse: _____

2. Squares are drawn on the sides of a right triangle. Find the area of the square on the hypotenuse, then the length of the hypotenuse.

(a) The squares on the legs have areas 49 cm^2 and 576 cm^2.

(b) The squares on the legs have areas 1.44 m^2 and 2.56 m^2.

3. Find the hypotenuse of each right triangle. (The answers are exact.)

(a) legs 16 cm and 30 cm (b) legs 2.8 m and 4.5 m (c) legs 33 ft and 56 ft

4. Find the hypotenuse of each right triangle, to the nearest tenth.

(a) legs 4 in and 7 in (b) legs 12.5 cm and 3.2 cm (c) legs 0.6 m and 0.9 m

5. Without a calculator, find c^2, then say which two whole numbers the hypotenuse c is between.

(a) legs 5 and 6 (b) legs 3 and 9

B. Applying

6. A computer monitor's screen is 23.5 in wide and 13.2 in high. Is it a 27-inch monitor? Show how you know.

7. A soccer field in Red Deer is 100 m long and 64 m wide. A player jogs from one corner to the opposite corner.

(a) How far is the jog straight across the field (the diagonal), to the nearest tenth of a metre?

(b) How much shorter is that than jogging along two sidelines?

8. A zipline starts on a platform 12 m higher than the end platform. The two platforms are 85 m apart, measured along the ground.

(a) About how long is the cable (pulled straight)? Round to the nearest tenth of a metre.

(b) Why is the cable only a little longer than 85 m?

9. A shed roof rafter rises 3 ft over a horizontal run of 6 ft 8 in. How long is the rafter, to the nearest inch? Will an 8 ft board be long enough?

10. Lee says, "$a^2 + b^2 = c^2$ works for any triangle." Use a triangle with all three sides 5 cm long (it has no right angle) to show that Lee is wrong.

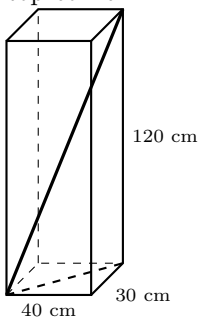
11. A TV screen is 32 in wide and 18 in high. Its size (the diagonal) is about

A. 14.0 in **B.** 36.7 in **C.** 50.0 in **D.** 1348 in

12. Numerical Response The foot of a ladder is 1.4 m from a wall, and the top of the ladder touches the wall 5.2 m above the ground. The length of the ladder, to the nearest tenth of a metre, is _____ m.

C. Stretch

13. A box is 40 cm long, 30 cm wide and 120 cm tall. Will a 125 cm hockey stick fit inside, from a bottom corner to the opposite top corner?



Hint: first find the diagonal across the bottom of the box (dashed). Then use it as a leg of a second right triangle that stands up inside the box.

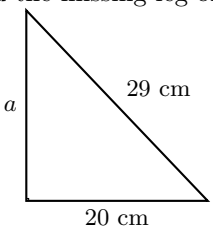
14. On a trail map of Kananaskis, a lookout is 2.5 km east and 3.5 km north of a campground. How far is the lookout from the campground in a straight line, to the nearest tenth of a kilometre?

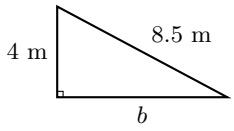
15. Which has the longer diagonal: a rectangle 5 m by 12 m, or a square 9 m by 9 m? Show your work.

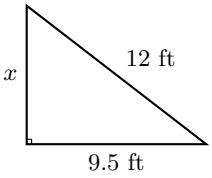
Extra Practice 3: The Pythagorean Theorem: Finding a Leg, and Squaring a Corner

A. Skills

1. Find the missing leg of each right triangle. Round to the nearest tenth if needed.

(a) 

(b) 

(c) 

2. Each row describes a right triangle. Write **leg** or **hypotenuse** for the missing side, then find it to the nearest tenth.

What you know	**Missing side is a**	**Work and answer**
(a) legs 1.5 m and 3.6 m		
(b) hypotenuse 26 in, a leg 24 in		
(c) a leg 11 cm, hypotenuse 14 cm		

3. Is each triangle a right triangle? Show your work.

(a) 11 m, 60 m, 61 m (b) 6 cm, 7 cm, 9 cm (c) 4.1 ft, 0.9 ft, 4.0 ft

4. In $\triangle DEF$ the right angle is at E. The sides are d, e and f (each across from the matching corner).

(a) Which side is the hypotenuse? Write the Pythagorean theorem for $\triangle DEF$.

(b) Rearrange your formula to find d, and again to find f.

5. Mia had to find a leg when the hypotenuse is 15 cm and the other leg is 9 cm. She wrote $b^2 = 9^2 - 15^2 = -144$ and said the triangle can't exist. What did she do wrong? Find the leg.

B. Applying

6. A laptop screen is 15.6 in on the diagonal and 13.6 in wide. How tall is the screen, to the nearest tenth of an inch?

7. A wheelchair ramp at a community hall in Brooks is 4.82 m long. It covers 4.8 m along the ground.

(a) How high does the ramp rise, to the nearest hundredth of a metre?

(b) A common guideline says a ramp should rise no more than about 1 m for every 12 m along the ground. Does this ramp follow it? Explain.

8. A softball diamond is a square with 60 ft between the bases. How far does the catcher throw from home plate to second base, to the nearest tenth of a foot?

9. A concrete crew builds the wooden form for a rectangular pad 3.6 m by 4.8 m.

(a) How long should each diagonal be? Which 3-4-5 multiple is this?

(b) Why does the crew measure *both* diagonals, not just one?

10. A triangle has sides of 20 cm, 21 cm and 29 cm. Which statement is true?

A. It is a right triangle, because $20^2 + 21^2 = 29^2$.

B. It is not a right triangle, because $20 + 21$ is not 29.

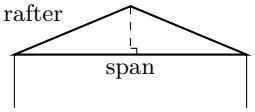
C. It is a right triangle, because 29 is the longest side.

D. It is not a right triangle, because $21^2 + 29^2$ is not 20^2.

11. Numerical Response A framer checks a picture frame that should be a rectangle 40 cm by 55 cm. If its corners are square, each diagonal, to the nearest tenth of a centimetre, is _____ cm.

C. Stretch

12. A house has a peaked roof. Each rafter runs from the peak down to the top of a side wall. The peak is straight above the middle of the house.



(a) The house is 9.6 m wide (the span) and the peak is 2.0 m above the top of the walls. How long is each rafter?

(b) On a garage, the span is 12 m and each rafter is 6.5 m long. How high is the peak above the top of the walls?

13. A crew is laying out a house foundation 30 ft by 40 ft. Their tape measure is only 25 ft long.

(a) If the corners are square, how long is the full diagonal of the foundation?

(b) Which 3-4-5 multiple should they use at each corner to check it with their tape? Explain.

14. Long ago, builders are said to have made square corners with a loop of rope tied with knots that split it into 12 equal sections.

(a) How can you stretch the rope into a triangle with a right angle? Explain why it works.

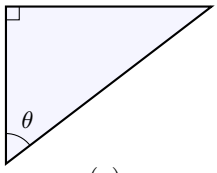
(b) Could a loop with 30 equal sections also make a right triangle using 5, 12 and 13 sections? Show your work.

Extra Practice 4: Similar Right Triangles and the Tangent Ratio

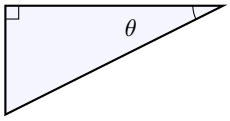
Use your calculator in DEG mode. Diagrams are not drawn to scale. Round lengths to the nearest tenth unless a question says otherwise.

A. Skills

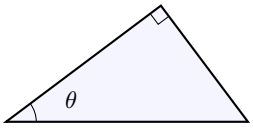
1. Write **opp**, **adj** or **hyp** beside each side. Use the angle marked θ.



(a)



(b)

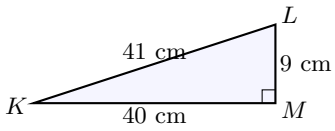


(c)

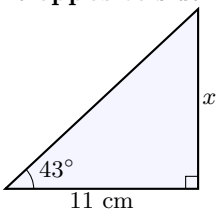
2. Find each value to 4 decimal places.

θ	5°	25°	52°	68°
$\tan\theta$				

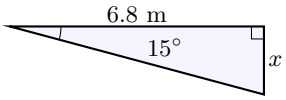
3. In $\triangle KLM$, $\angle M = 90°$, $KM = 40$ cm, $LM = 9$ cm and $KL = 41$ cm. Write $\tan K$ and $\tan L$ as fractions and as decimals (4 decimal places).



4. Find x. Here x is the **opposite** side.



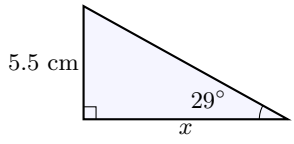
(a)



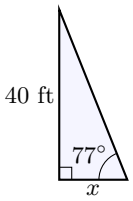
(b)

5. Find x. Here x is the **adjacent** side.

(a)

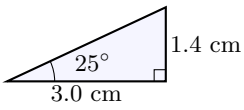


(b)

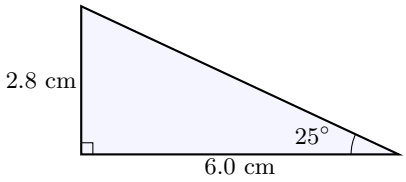


B. Applying

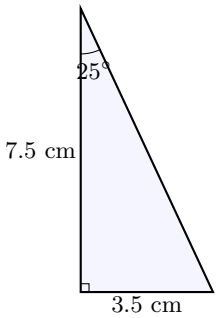
6. Three right triangles each have a 25° angle.



Triangle 1



Triangle 2



Triangle 3

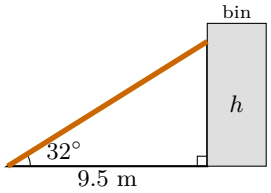
(a) Complete the table. Round opp ÷ adj to 2 decimal places.

	Triangle 1	**Triangle 2**	**Triangle 3**
opp (cm)			
adj (cm)			
opp ÷ adj			

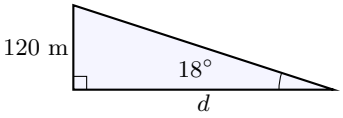
(b) Explain why the ratio is the same in all three triangles.

(c) A right triangle with a 25° angle has adj = 20 m. Use $\tan 25°$ to find opp.

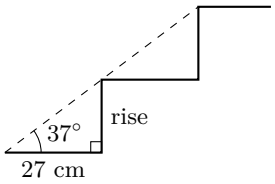
7. A grain auger leans against the wall of a grain bin. It makes a 32° angle with the ground, and its bottom end is 9.5 m from the bin wall. How high up the wall does the auger reach?



8. A ski run drops 120 m from top to bottom. The slope makes an 18° angle with the horizontal. How far does the run go horizontally?



9. A staircase rises at a 37° angle. Each step has a run (depth) of 27 cm. How tall is each step (the rise)?



10. Jaya checks her calculator and gets $\tan 45° = 1.619775191$. What is wrong?

A. Nothing: $\tan 45°$ is about 1.62.

B. The calculator is not in DEG mode.

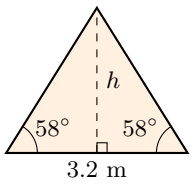
C. She should have pressed [2nd] before [tan].

D. 45° is too big an angle for the tangent.

11. Numerical Response A wheelchair ramp makes a 4.8° angle with the ground. It reaches 9.0 m along the ground. The height it rises, to the nearest hundredth of a metre, is _____ m.

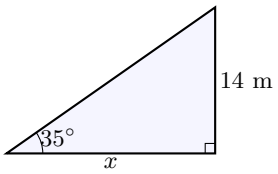
C. Stretch

12. The front of an A-frame tent is an isosceles triangle. It is 3.2 m wide at the bottom, and each side makes a 58° angle with the ground. How tall is the tent? *Hint:* the height cuts the triangle into two right triangles.



13. A carpenter says a roof with a pitch of 7/12 (it rises 7 for every 12 across) slopes at "about 30°". Use tangents to decide whether the roof is a little steeper or a little less steep than 30°.

14. Sam wants side x in this triangle. He writes:
$\tan 35^\circ = \dfrac{x}{14}$, *so* $x = 14 \times \tan 35^\circ = 9.8$ *m.*
(a) What mistake did Sam make? How could the 45° check have warned him?



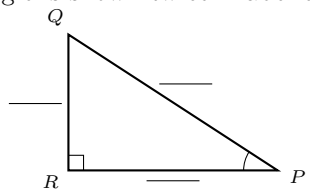
(b) Find the correct value of x.

Extra Practice 5: The Sine Ratio

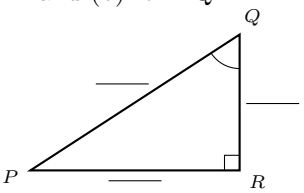
Use your calculator in DEG mode. Round lengths to the nearest tenth and sine values to 4 decimal places unless a question says otherwise. $\sin\theta = \dfrac{\text{opp}}{\text{hyp}}$

A. Skills

1. The same triangle is shown twice. Label each side opp, adj or hyp, (a) for $\angle P$ and (b) for $\angle Q$.

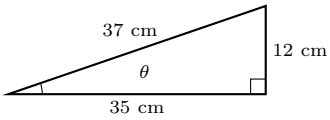


(a) For $\angle P$

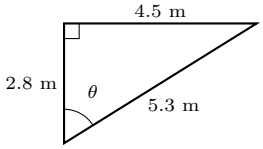


(b) For $\angle Q$

2. Write $\sin\theta$ as a fraction and as a decimal.



(a)

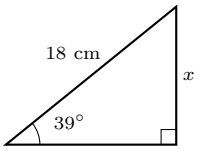


(b)

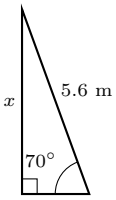
3. Find each value to 4 decimal places. Then list the four angles in order from the smallest sine to the largest.

(a) $\sin 64° =$ _____ (b) $\sin 8° =$ _____ (c) $\sin 86° =$ _____ (d) $\sin 41° =$ _____

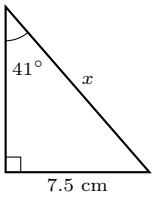
4. Find x.



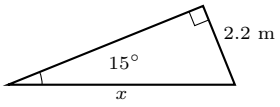
(a)



(b)



(c)



(d) (not to scale)

B. Applying

5. A chairlift cable at a ski hill in the Rockies is 1450 m long. It climbs at an average angle of 21°. Sketch a right triangle. How much higher is the top station than the bottom station?

6. An escalator in a Calgary mall climbs 5.4 m from one floor to the next. It makes a 30° angle with the floor. How long is the escalator? Find the answer in your head first using $\sin 30° = 0.5$, then check with your calculator.

7. A survey drone flies 150 m in a straight line from its launch pad, climbing at 12° above the horizontal. How high above the launch pad is it?

8. The deck of a tilting tow truck tips down until its back end touches the road. The front end of the deck is then 1.6 m above the road, and the deck makes an 11° angle with the road. How long is the deck?

9. A right triangle has a hypotenuse of 10 cm and a 53° angle. Which is the length of the side opposite the 53° angle, to the nearest tenth?

A. 6.0 cm **B.** 8.0 cm **C.** 12.5 cm **D.** 0.8 cm

10. Numerical Response A fire truck's ladder is 30.5 m long. It is raised to 64° from the horizontal, and its bottom end turns on a pivot 2.8 m above the ground. The height of the top of the ladder above the ground, to the nearest tenth of a metre, is _____ m.

C. Stretch

11. A bush plane leaves Fort McMurray and flies 60 km in a straight line. Its path makes a 25° angle with the line pointing due north, toward the east. Sketch a right triangle with one leg pointing north and one pointing east. How far east of Fort McMurray is the plane? Explain why the sine ratio works here.

12. A right triangle has an opposite side of 9.0 cm and a 37° angle.

(a) Find the hypotenuse.

(b) Use the Pythagorean theorem to find the third side. Use the full calculator value from (a).

13. Arjun has a 4.0 m ladder. He needs to reach a window ledge 3.7 m above the ground, with the ladder at the safe angle of 75°.

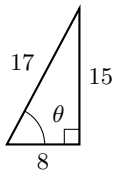
(a) Will his ladder reach?

(b) What is the shortest ladder that would reach the ledge at 75°? Give the answer to the nearest tenth of a metre, and explain how you rounded.

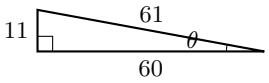
Extra Practice 6: The Cosine Ratio

A. Skills

1. Write $\cos\theta$ as a fraction. Then write it as a decimal to 4 decimal places.



(a)



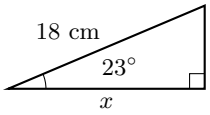
(b)

2. Use your calculator. Round to 4 decimal places.

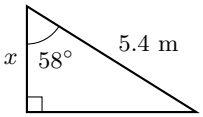
(a) $\cos 5° =$ _____ (b) $\cos 38° =$ _____ (c) $\cos 77° =$ _____ (d) $\sin 52° =$ _____

(e) Why are the answers to (b) and (d) the same?

3. Find x. Round to the nearest tenth.

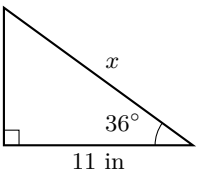


(a)

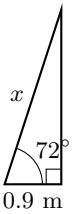


(b)

4. Find x. Round to the nearest tenth.

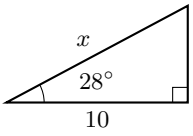


(a)

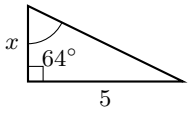


(b)

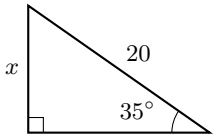
5. For each triangle, choose sin, cos or tan to find x. Then find x to the nearest tenth.



(a)



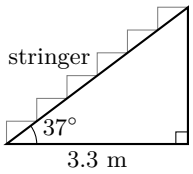
(b)



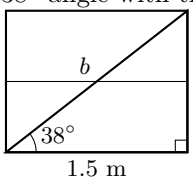
(c)

B. Applying

6. A carpenter is building stairs. The stringer (the sloping board that holds the steps) makes a 37° angle with the floor and covers a horizontal run of 3.3 m. How long is the stringer, to the nearest tenth of a metre?



7. A wooden gate is a rectangle 1.5 m wide. A diagonal brace b goes from one bottom corner to the opposite top corner. It makes a 38° angle with the bottom rail. How long is the brace, to the nearest tenth of a metre?



8. A drone flies 260 m in a straight line, climbing at 9° to the horizontal. Sketch the triangle. How far does it travel horizontally, to the nearest tenth of a metre?

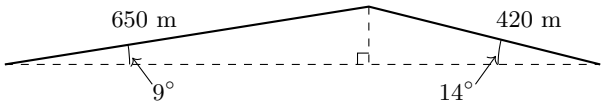
9. The bottom of a ladder is 1.5 m from a wall. The ladder makes a 75° angle with the ground. Which is the length of the ladder?

A. 0.4 m **B.** 1.6 m **C.** 5.8 m **D.** 1.4 m

10. Numerical Response A conveyor belt at a gravel pit is 24 m long and slopes up at 21° to the ground. The horizontal distance it covers is _____ m, to the nearest tenth of a metre.

C. Stretch

11. A cross-country ski trail climbs a straight slope at 9° for 650 m (measured along the snow). Then it goes down the other side at 14° for 420 m. What horizontal distance does the trail cover? Round to the nearest metre.



12. Jordan says, "My triangle has an adjacent side of 5 cm and a hypotenuse of 4 cm, so $\cos\theta = 1.25$." Explain why this cannot be true.

13. In $\triangle ABC$, $\angle C = 90°$, $a = 33$, $b = 56$ and $c = 65$.

(a) Write $\cos A$ and $\sin B$ as fractions. What do you notice?

(b) $\cos 50° \approx 0.6428$. Without a calculator, what is $\sin 40°$? Explain.

Extra Practice 7: Angles of Elevation and Depression

A. Skills

1. Write **elevation** or **depression** for the angle in each situation.

(a) A climber at the top of a rock wall looks down at her partner on the ground. _____

(b) A hockey fan looks up at the scoreboard. _____

(c) A pilot looks down at a lake. _____

(d) A child at the bottom of a slide looks up at the top. _____

2. Find each angle.

(a) From a roof, the angle of depression to a dog in the yard is 33°. The angle of elevation from the dog to the roof is _____.

(b) Sam looks down from a tower. His line of sight makes an angle of 80° with the vertical. The angle of depression is _____.

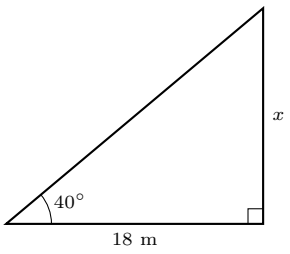
(c) From a boat, the angle of elevation to the top of a cliff is 47°. The angle of depression from the top of the cliff to the boat is _____.

3. Find the angle of elevation for each clinometer reading.

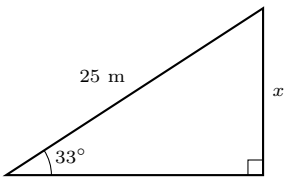
Reading	75°	49°	117°	140°	83°
Angle of elevation					

4. Each triangle has an angle of elevation at the lower left. Find x to the nearest tenth. Write the ratio you use.

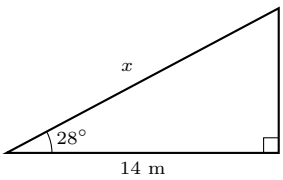
(a)



(b)



(c)



B. Applying

5. Kofi stands 40 m from a grain elevator near Lethbridge. His eyes are 1.5 m above the ground. The angle of elevation to the top of the elevator is 37°. How tall is the grain elevator, to the nearest tenth of a metre?

6. Rosa flies a drone. It is 120 m from her controller along her line of sight, at an angle of elevation of 26°. She holds the controller 1.4 m above the ground. How high is the drone above the ground, to the nearest tenth of a metre?

7. From a bridge deck 18 m above the Bow River, the angle of depression to a kayak is 24°. How far is the kayak from the point on the water right under the bridge, to the nearest tenth of a metre?

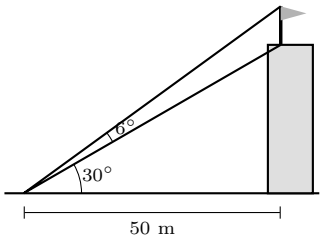
8. From a point 25 m from a light standard, the angle of elevation to its top is 20°. Your eyes are 1.5 m above the ground. How tall is the light standard?

A. 9.1 m **B.** 10.6 m **C.** 70.2 m **D.** 10.1 m

9. Numerical Response A ski run goes down in a straight line at an angle of depression of 15° from the top. A skier travels 650 m along the run. How many metres lower is she, to the nearest metre? _____ m

C. Stretch

10. A flagpole stands on the roof of a building. From a point on the ground 50 m from the building, the angle of elevation to the bottom of the flagpole is 30° and to the top of the flagpole is 36°. How long is the flagpole itself, to the nearest tenth of a metre?



11. A plane is 1200 m above the runway. It comes in to land at an angle of depression of 3°. Round to the nearest tenth of a kilometre.

(a) Find the horizontal distance to the touchdown spot.

(b) Find the distance the plane flies along its path.

(c) Your two answers are almost the same. Why does that make sense?

12. Owen stands 20 m from an apartment building. His eyes are 1.6 m above the ground, and the angle of elevation to the roof is 50°. He says the roof is 15.3 m above his eyes.

(a) Use the 45° idea to explain why his answer is not reasonable.

(b) What mistake did he make? Find the correct height of the building to the nearest tenth.

13. From the top of a 60 m communication tower, a worker sees two trucks on opposite sides of the tower, on level ground. The angle of depression to one truck is 20° and to the other is 35°. How far apart are the trucks, to the nearest tenth of a metre? Sketch it first.

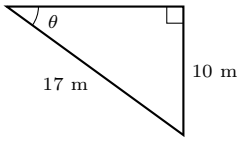
Extra Practice 8: Finding Angles with Inverse Trig Ratios

A. Skills

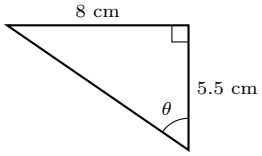
1. Find each angle θ to the nearest degree.
(a) $\sin\theta = 0.25$ (b) $\cos\theta = 0.6$ (c) $\tan\theta = 1.8$ (d) $\cos\theta = 0.95$

2. Find each angle to the nearest degree.
(a) $\sin E = \frac{4}{9}$ (b) $\cos F = \frac{13}{20}$ (c) $\tan G = \frac{17}{6}$

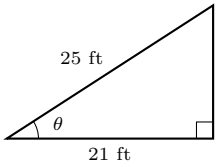
3. Label the two given sides opp, adj or hyp from θ. Then find θ to the nearest degree.



(a)



(b)



(c)

4. One acute angle of a right triangle is given. Find the other acute angle.
(a) 23° _____ (b) 58.6° _____

5. In $\triangle DEF$, $\angle F = 90°$, $DF = 21$ cm, $EF = 20$ cm and $DE = 29$ cm.
(a) Find $\angle D$ with the tangent ratio.

(b) Check $\angle D$ with the sine ratio.

(c) Find $\angle E$.

B. Applying

6. A grain auger is 40 ft long. Its top end rests on the edge of a bin 22 ft above the ground. What angle does the auger make with the ground?

7. A 16 ft ladder leans against a barn. Its foot is 5 ft from the wall.
(a) Find the angle between the ladder and the ground.

(b) Is the ladder set up safely (about 75°)? What should change?

8. Two staircases are being compared. Stair A: each step rises 19 cm and goes 25 cm forward. Stair B: each step rises 17 cm and goes 29 cm forward. Find the angle of each staircase with the floor. Which is steeper?

9. A playground slide is 3.4 m long. The top of the slide is 1.8 m above the ground. What angle does the slide make with the ground?

10. A plane is landing at Calgary's airport. It is 150 m above the ground and 2.9 km (measured along the ground) from the point where it will touch down. Find the angle of descent (the angle of depression from the plane to that point), to the nearest tenth of a degree.

11. A rafter is 4.8 m long and rises 1.6 m from the wall to the peak. To the nearest degree, the angle between the rafter and the horizontal is

A. 18° **B.** 19° **C.** 71° **D.** 72°

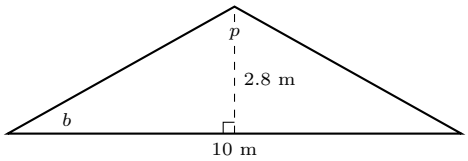
12. Numerical Response A wheelchair ramp rises 0.4 m over a horizontal distance of 5 m. The angle of the ramp, to the nearest tenth of a degree, is _____°.

C. Stretch

13. A homeowner wants a roof that makes an angle of about 30° with the horizontal. Roofers describe pitch as "rise/12".
(a) What rise (to the nearest whole inch) gives about 30°?

(b) Check your pitch with an inverse trig ratio.

14. A roof truss is an isosceles triangle. It is 10 m wide at the bottom and its peak is 2.8 m above the bottom, in the middle.



(a) Find the base angle b.

(b) Find the angle p at the peak.

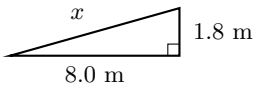
15. Noah says: "If I make both legs of a right triangle twice as long, the angles will double too." Test his claim with legs 5 cm and 7 cm, then legs 10 cm and 14 cm. Is he right? Explain.

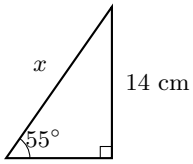
Extra Practice 9: Solving Right Triangles: Choosing the Right Ratio

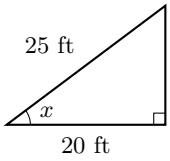
Round sides to the nearest tenth and angles to the nearest degree unless the question says otherwise. Your calculator must be in DEG mode.

A. Skills

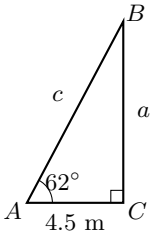
1. Name the tool and write the equation. Then find x.

(a) 

(b) 

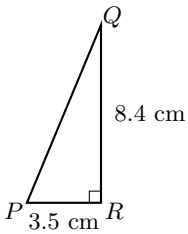
(c) 

2. Solve $\triangle ABC$, where $\angle C = 90°$, $\angle A = 62°$ and $b = 4.5$ m.



3. Solve $\triangle ABC$, where $\angle C = 90°$, $a = 13$ ft and the hypotenuse $c = 21$ ft.

4. Solve $\triangle PQR$, where $\angle R = 90°$, $PR = 3.5$ cm and $QR = 8.4$ cm.



5. Ari solved a right triangle with $\angle A = 30°$ and a hypotenuse of 10 cm. He wrote: $a = 8.7$ cm (opposite $\angle A$) and $b = 5.0$ cm. How can you tell he made a mistake? Fix it.

B. Applying

6. A 35 m guy wire is fastened to a cell tower 30 m above the ground.

(a) What angle does the wire make with the ground?

(b) How far from the base of the tower is the wire anchored?

7. A plane flying at 1200 m starts a straight descent to the runway at Calgary's airport at an angle of depression of 3°.

(a) How far does the plane travel along the ground before it lands? Give your answer in km, to the nearest tenth.

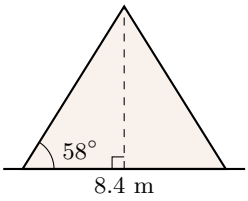
(b) How far does it fly along its slanted path?

(c) Why are your answers so close?

8. An A-frame cabin near Canmore is 8.4 m wide at the floor. Each side of the roof makes an angle of 58° with the floor, and the sides meet at the peak.

(a) How tall is the cabin at the centre?

(b) How long is each sloped side?



9. A new ramp at a rink in Brooks is 6.0 m long and rises 0.55 m.

(a) Find the angle of the ramp, to the nearest tenth of a degree.

(b) Is it within the guidance of about 4.8° or less? If not, how long should a ramp rising 0.55 m at 4.8° be?

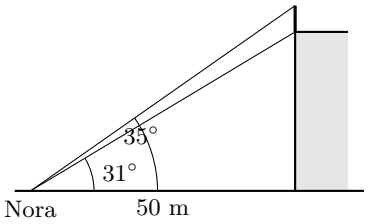
10. In a right triangle, the side opposite θ is 7 cm and the side adjacent to θ is 10 cm. Which gives θ?

A. $\sin^{-1}\left(\frac{7}{10}\right)$ **B.** $\tan^{-1}\left(\frac{7}{10}\right)$ **C.** $\tan^{-1}\left(\frac{10}{7}\right)$ **D.** $\tan\left(\frac{7}{10}\right)$

11. Numerical Response A ski run at Nakiska drops 240 m vertically over a slope length of 950 m. The average angle of the slope with the horizontal, to the nearest degree, is _____°.

C. Stretch

12. Nora stands 50 m from an office building in Edmonton. From the ground, the angle of elevation to the bottom of a flagpole on the roof is 31°, and to the top of the flagpole is 35°. How tall is the flagpole?



13. A moving box is 60 cm long, 40 cm wide and 45 cm tall. Liam wants to fit a curtain rod in it, corner to corner.

(a) Find the diagonal of the bottom of the box.

(b) The rod goes from a bottom corner to the opposite top corner. It makes a right triangle with the bottom diagonal and a 45 cm edge. What is the longest rod that fits?

(c) What angle does that rod make with the bottom of the box?

Extra Practice 10: Unit 7 Review

Practice Test 2. Treat this like the unit test. Show your work, and use your calculator in DEG mode. Round lengths to the nearest tenth and angles to the nearest degree unless a question says otherwise.

A. Skills

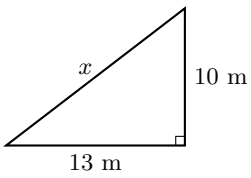
1. Fill in the blanks.

(a) $15^2 =$ _____ (b) $\sqrt{289} =$ _____ (c) $\sqrt{45} \approx$ _____

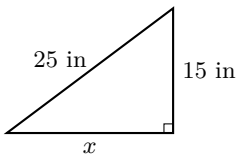
(d) A right triangle has an acute angle of 58°. Its other acute angle is _____.

2. Find each side marked x with the Pythagorean theorem.

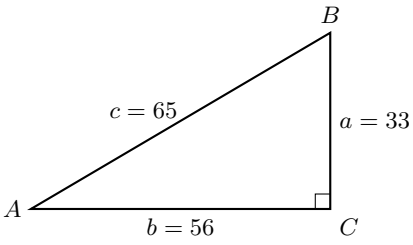
(a)



(b)



3. In $\triangle ABC$, $\angle C = 90°$, $a = 33$, $b = 56$ and $c = 65$.



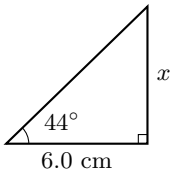
(a) For $\angle A$, name opp, adj and hyp.

(b) Write $\sin A$, $\cos A$ and $\tan A$ as fractions and as decimals to 4 decimal places.

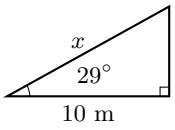
(c) Which side is opposite $\angle B$? Write $\sin B$.

4. Find each side marked x.

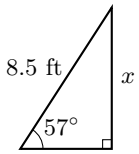
(a)



(b)

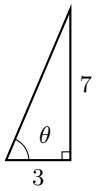


(c)

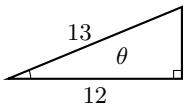


5. Find each angle θ.

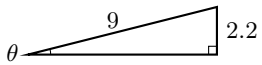
(a)



(b)



(c)



B. Applying

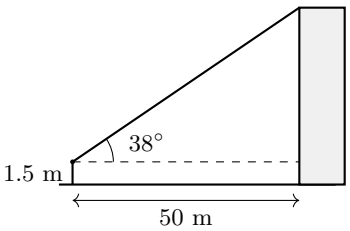
6. A carpenter cuts two triangular corner shelves. Which one has a square corner? Show your work.

(a) sides 45 cm, 60 cm and 75 cm

(b) sides 40 cm, 50 cm and 66 cm

7. Not counting the end zones, a Canadian football field is a rectangle 110 yd long and 65 yd wide. A player runs straight from one corner of it to the opposite corner. How far does he run?

8. A surveyor sets up her instrument 50 m from the base of an office building. The instrument is 1.5 m above the ground, and the angle of elevation to the top of the building is 38°. How tall is the building?



9. A ramp to the front door of a clinic must rise 0.75 m. The town asks for a slope of 1 : 12 (1 m up for every 12 m along the ground).

(a) How long must the horizontal run be?

(b) How long is the sloped surface of the ramp?

(c) Find the angle of the ramp to the nearest tenth of a degree.

10. At 4:00 one afternoon, Leila, who is 1.8 m tall, casts a shadow 2.4 m long. At the same time, a lamp post casts a shadow 10 m long.

(a) Find the angle of elevation of the sun.

(b) The sun's rays are parallel, so the lamp post's triangle has the same angle and the same tangent ratio. Find the height of the lamp post.

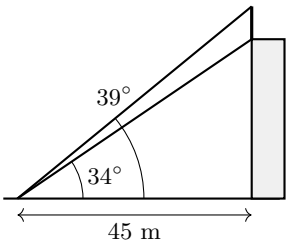
11. A right triangle has a 32° angle, and the side opposite it is 9 m. Which equation gives the hypotenuse, x?

A. $x = 9 \times \sin 32°$ **B.** $x = 9 \div \sin 32°$ **C.** $x = 9 \div \cos 32°$ **D.** $x = \sin 32° \div 9$

12. Numerical Response A rope runs from the top of a 2.4 m tent pole to a peg in the ground and makes an angle of 40° with the ground. To the nearest tenth of a metre, the rope is _____ m long.

C. Stretch

13. An antenna stands on top of a hotel. From a point on the ground 45 m from the hotel, the angle of elevation to the top of the hotel is 34° and to the top of the antenna is 39°. How tall is the antenna?



14. A car trunk is shaped like a box 1.2 m wide, 0.8 m deep and 0.6 m tall. Zara wants to fit in a hockey stick 1.5 m long.

(a) Find the diagonal of the trunk's floor. Will the stick lie flat on the floor?

(b) The longest straight line in the trunk goes from a bottom corner to the opposite top corner. It is the hypotenuse of a right triangle with legs equal to the floor diagonal and the height. Will the stick fit that way?

15. A right triangle has an adjacent side of 9 cm and a hypotenuse of 15 cm. Reza wrote $\cos\theta = \frac{15}{9}$. He pressed the keys below, and the calculator showed an error.

[2nd] [cos] 15 [÷] 9 [)] [ENTER]

(a) Explain why the calculator gave an error.

(b) Fix Reza's work and find θ. Then check with the other acute angle.